ZIDAN FERDIANSYAH

📞 +62 81319535441 | 📧 zidanferdi13@gmail.com | 🌐 linkedin.com/in/zidanferdiansyah
🏠 Pondok Ungu Permai Blok AL 8/9, Bekasi Regency

PROFESSIONAL SUMMARY

IT System & IoT Engineer with extensive hands-on experience in **industrial automation, manufacturing digitalization, and large-scale IoT deployments** across mining and factory environments. Currently responsible as **Lead Device Display & Project PIC**, managing end-to-end development of **24/7 monitoring systems** for heavy equipment and industrial machinery.

Proven expertise in **integrating PLC systems, sensors, CAN Bus, and VHMS data** into reliable IT infrastructures using **MQTT, Node.js, real-time dashboards, and web-based monitoring systems**. Strong background in **system uptime assurance, data accuracy, and operational reliability**, with experience deploying solutions from **edge devices (PLC, Raspberry Pi, gateways)** to **centralized servers and cloud-connected systems**.

Demonstrated ability to **design, implement, and maintain full-stack industrial systems**, bridging OT and IT to support data-driven operations, predictive monitoring, and continuous improvement in large-scale industrial environments.

CORE COMPETENCIES

• Industrial IoT & Manufacturing Automation
• PLC, Sensor & Weighing System Integration
• IT System Infrastructure & Edge Computing
• MQTT, Socket.IO & Real-Time Data Streaming
• Node.js Backend & API Development
• Database Management (MongoDB, MariaDB)
• Production & Utility Monitoring Systems
• Dashboard & Data Visualization (Power BI, Node-RED)
• Field Deployment, Calibration & Troubleshooting

EDUCATION

Lampung University – Indonesia
Bachelor's Degree in Agricultural Engineering | GPA: 3.77/4.00
Aug 2019 – Jun 2023
Thesis: Design and Development of Automatic Measuring Device for Cassava Starch using Arduino UNO and Load Cell.

WORK EXPERIENCE

IT & IIoT Engineer – PT Anugerah Mortar Abadi
Nov 2025 – Present

- Lead end-to-end development of industrial monitoring systems for **manufacturing operations, integrating PLCs, load cells, and production equipment** into centralized dashboards.
- Act as PIC for factory-wide **device monitoring systems, ensuring data accuracy, system reliability, and continuous operation** across multiple sites.
- Digitized manual production and weighing processes by connecting **PLC-based weighing systems to web-based monitoring** platforms, significantly reducing manual reporting errors.
- Designed and maintained local **server infrastructure, MQTT brokers, and Node.js**-based backend systems for real-time factory data acquisition.
- Integrated **CAN Bus, MOD Bus, and heavy equipment health data** into centralized monitoring systems to support preventive maintenance and operational decision-making.
- Collaborated closely with production, maintenance, and management teams to align IT systems with manufacturing workflows and operational needs.

IoT Engineer – PT Pamapersada Nusantara
Mar 2024 – Present

- Deployed and maintained over **400 IoT devices** in coal mining operations, achieving **98% system uptime** through proactive diagnostics and field inspections.
- Conducted real-time fatigue detection for **400+ operators** using AI-based camera systems, contributing to a **25% reduction in fatigue-related incidents**.
- Integrated sensors, AI cameras, and measuring tools with cloud infrastructure using **MQTT and Socket.IO**, enabling seamless real-time data transmission.
- Developed automated dashboards using **Power BI** to visualize operational metrics, supporting **data-driven planning and resource allocation**.
- Led the troubleshooting and calibration of IoT hardware, reducing maintenance response time by **30%**.
- Collaborated with cross-functional teams (engineering, safety, IT) to ensure that deployed systems met both operational and compliance standards.

Robotics Instructor – BrightCHAMPS, India
Oct 2023 – Mar 2024

- Delivered robotics and coding instruction to over **100 students** (Grades 3–10) across global time zones, using a combination of Arduino-based hardware and interactive software.
- Designed and facilitated **project-based learning modules** for game development, app prototyping, and website creation, increasing student engagement and project completion rates.

- Instructed students in **logical programming fundamentals** and microcontroller operations, including sensors, actuators, and electrical systems.
- Adapted teaching strategies to suit international learners, ensuring a high satisfaction rate and improved technical comprehension among non-native English speakers.

Agricultural Extension Officer – Dinas Pertanian Bandar Lampung
Jun 2022 – Aug 2022

- Delivered technical consultations to **20+ local farmers**, offering actionable solutions for land-use and crop management challenges.
- Conducted hands-on training sessions on modern **agricultural technologies** and **eco-friendly** practices to improve crop productivity.
- Designed customized assistance plans based on village-level agricultural needs assessments.
- Organized and executed field surveys and inspections, ensuring alignment with planned agricultural interventions and validating reported outcomes.

ORGANIZATIONAL EXPERIENCE

Head of Community Service – Agricultural Engineering Student Association
Feb 2022 – Jan 2023

- Led 12-member team in executing 2 major village development programs, impacting 200+ residents through technology transfer and agricultural training.
- Established sustainable partnerships with 2 rural villages, resulting in ongoing collaboration programs.

Member of Community Service Division
Feb 2021 – Jan 2022

- Played a key role in planning and documenting village empowerment results, supporting proposal approvals and university reporting.
- Liaised with village officials to maintain progress tracking in accordance with MoU.

PROJECTS

IoT-based Smart Farming System – Lampung University
Jan 2023 – Aug 2023

- Enabled autonomous environmental monitoring with 6 sensors powered by solar panels, integrated with Node-RED for control flow and real-time data to Google Sheets.
- System tested in greenhouse setup, improving decision speed for irrigation and fertilizer application by 40%.

Cassava Starch Measuring Device – Lampung University
Sep 2022 – Mar 2023

- Achieved 95% measurement accuracy using load cell–based specific gravity detection, reducing manual labor in starch processing.
- The tool reduced processing time by 40%, validated in lab-scale and student demonstration projects.

CERTIFICATIONS

- Basic SCADA Using Siemens WinCC Explorer – Anak Teknik Indonesia (2023)
- Introduction to IoT: Exploring IoT Engineer – Edspert.id (2023)
- KMPD & KMKOP – PT Berau Coal Energy (2024)